# AUDITRA

## AI-POWERED BRAND GOVERNANCE & CONTENT REVIEW PLATFORM



## 1. Executive Summary & Mission

AUDITRA is a pre-publication brand-governance control layer that evaluates submitted marketing and corporate content against an organization's approved brand rules, terminology, facts, claims, and compliance disclosures. Unlike generic LLM grammar checkers or writing chatbots, AUDITRA's core differentiator is **evidence-first, human-controlled brand review**. Every single finding provides traceable citation of the exact violated rule, match offsets, explanation, and targeted one-click corrections.

## 2. System Architecture & Technology Stack

| Layer | Component & Implementation | Governance Function |
|---|---|---|
| Frontend | React 19, TypeScript, Vite, Tailwind CSS, Lucide Icons | Tri-pane Review Studio, Rulebook Manager, Command Centre, Audit Trail |
| Backend | Python 3.12, FastAPI, Pydantic v2 schemas | High-throughput REST API, validation, error boundaries, CORS handling |
| Persistence | SQLite with Write-Ahead Logging (WAL) & SQLAlchemy ORM | Durable relational persistence for versions, findings, decisions & audit events |
| Compliance Engine | Deterministic Regex & Rule Matcher (Zero-LLM dependency) | Exact/case-insensitive phrase detection, preferred terms, required disclosures |
| AI Assistant | Server-side LLM (Gemini/OpenAI/Claude) + Honest Fallback | Tone analysis & claim grounding with strict prompt-injection protection |
| State Machine | Server-enforced transition validation: Draft -> Review -> Approved | Prevents unauthorized state jumps; guarantees human oversight |

## 3. Core Implemented Features & Verification

| Feature Area | Implementation Status | Verified Capability |
|---|---|---|
| First-Run & Sample | Implemented & Tested | Auto-seeding of 'Northstar' fictional brand with 7 realistic rules & 3 claims |
| Brand Rulebook | Implemented & Tested | Full CRUD, category filtering, severity tagging, and text/markdown guideline ingestion |
| Review Studio | Implemented & Tested | Live character counts, version checkpoints, risk score gauge (0-100), finding filters |
| Deterministic Engine | Implemented & Tested | Exact prohibited phrases, preferred terms, missing mandatory disclosures, unverified claims |
| Targeted Corrections | Implemented & Tested | One-click suggestion application, in-place text replacement, new version creation |
| Approval Workflow | Implemented & Tested | Server-validated state machine: Draft -> In Review -> Changes Requested / Approved / Rejected |
| Audit Trail | Implemented & Tested | Append-only event log capturing every edit, review run, suggestion, comment, and decision |
| Dashboard & Insights | Implemented & Tested | Aggregated metrics, severity distribution, status breakdown, recurring issue analytics |

## 4. Automated Testing & Reliability Evidence

A comprehensive automated test suite covers the entire critical workflow across 9 test suites with **100% pass rate** in 1.00s:
• **Health Endpoint**: Verifies API liveliness, database configuration, and AI engine status.
• **Deterministic Rule Engine**: Tested against prohibited phrases, preferred terms, required disclosures, unverified claims, and negative cases (clean copy producing 0 false positives).
• **Sample Brand Persistence**: Confirms SQLite WAL durability and instant workspace bootstrapping.
• **End-to-End Content Lifecycle**: Verifies draft creation -> compliance review -> targeted correction (v1->v2) -> submission -> reviewer approval -> audit trail verification.
• **State Machine Guardrails**: Proves illegal state transitions (e.g. Draft -> Approved directly) are strictly blocked with HTTP 400.
• **Guideline File Ingestion**: Verifies document parsing and rule extraction from markdown/text files.

## 5. Proposed Round 3 Strategic Roadmap

**Improvement 1: Multimodal Brand Governance**
Extend AUDITRA beyond text to cross-modal content governance. Inspect marketing creatives, banners, and video thumbnails for brand-palette compliance, logo exclusion zones, contrast accessibility, and embedded image OCR text verification.

**Improvement 2: Evidence-Grounded Brand Knowledge & Claim Verification**
Incorporate high-precision semantic retrieval (RAG) over authoritative internal documents (10-K filings, SOC2 audits, lab whitepapers) with automated provenance tracking and freshness expiration dates to prevent out-of-date product metrics.

**Improvement 3: Enterprise Governance & Publishing Integrations**
Introduce multi-tenant organizational hierarchies, SSO/SAML 2.0, dual-custody approval gates for high-risk legal claims, and publishing connectors for HubSpot, WordPress, and social media scheduling engines.

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